

Effect of inositol hexaphosphate (IP(6)) on human normal and leukaemic haematopoietic cells.



Inositol hexaphosphate (IP(6)), a naturally polyphosphorylated carbohydrate, has been reported to have significant in vivo and in vitro anticancer activity against numerous tumours, such as colon, prostate, breast, liver and rhabdomyosarcomas. To confirm this activity in haematological malignancies and to characterize some of the mechanisms of IP(6) action, we analysed its effects on human leukaemic cell lines and fresh chronic myelogenous leukaemia (CML) progenitor cells using a combined cellular and molecular approach. IP(6) had a dose-dependent cytotoxic effect on all of the evaluated cell lines, with accumulation in the G2M phase in two out of five cell lines tested. At the molecular level, cDNA microarray analysis after IP(6) exposure showed an extensive downmodulation of genes involved in transcription and cell cycle regulation and a coherent upregulation of cell cycle inhibitors. Furthermore, IP(6) treatment of fresh leukaemic samples of bone marrow CD34⁺ CML progenitor cells significantly inhibited granulocyte-macrophage colony-forming unit (CFU-GM) formation ($P = 0.0062$) in comparison to normal bone marrow specimens, which were not affected. No differentiating effect on HL60 cells was observed. Taken together, our results confirm the antiproliferative activity of IP(6) and suggest that it may have a specific antitumour effect also in chronic myeloid leukaemias, via active gene modulation.